



Empirical Relationship Structure in U.S. Macroeconomic Data for CHronoZ Benchmark Design

Correlation, Redundancy, Lead-Lag, and Rolling-Stability Evidence from a 36-Series FRED Universe

QuantFun Technologies - CHronoZ Economic Benchmark Modelling | Research Note v0.1 | 18 September 2026

Repository snapshot: e2153e19ea20 | <https://github.com/Rizuqura/CHronoZ-economicBenchmarkModelling>

Abstract

This study documents the first empirical relationship-mapping stage of the CHronoZ economic benchmark project. The purpose is not to forecast assets or declare causal economic laws, but to establish a reproducible statistical map of a U.S. macroeconomic data universe before benchmark, regime, and machine-learning layers are introduced. The underlying repository contains 36 cleaned FRED series spanning activity, labor, inflation, credit, policy/liquidity, and external/structural dimensions. A monthly analytical core of 27 series is examined using raw-level diagnostics, explicitly registered one-month transformations, Pearson and Spearman correlations, redundancy screening, calendar-aligned lead-lag scans, and 60-month rolling correlations. The strongest contemporaneous relationships concentrate in industrial activity, consumption, and inflation measures. Seven redundancy candidates survive a joint Pearson-Spearman screen, including an industrial triangle formed by Industrial Production, Manufacturing Production, and Capacity Utilization. Lead-lag scanning evaluates 8,775 pair-lag combinations: 194 pairs peak at zero lag and 157 at nonzero lag, but the strongest relationships remain mostly contemporaneous and nonzero peaks are treated as descriptive timing associations rather than predictive evidence. Rolling analysis produces 189,286 estimates and shows a critical distinction between structurally stable relationships and state-dependent relationships. Industrial Production-Capacity Utilization remains exceptionally stable, while Payroll-Unemployment and C&I Loans-Industrial Production vary materially through time. The central design implication is that CHronoZ should preserve variable-level benchmark states as canonical objects, treat redundancy clusters as aggregation controls rather than data deletions, and model relationship stability and anomaly as a separate evidence layer. These findings provide a quantitative foundation for later benchmark calibration, historical-analogue retrieval, LLM interpretation, machine-learning features, and downstream pricing and decision systems.

Keywords: *macroeconomic benchmark; correlation; redundancy; lead-lag; rolling correlation; economic state; CHronoZ; agentic research; FRED*

1. Research Objective and Scope

CHronoZ is being developed as an empirical macro-intelligence system in which the quantitative layer measures economic conditions and the reasoning layer interprets those measurements. The immediate research problem is therefore upstream of forecasting: before a benchmark can classify a variable as above, below, or near neutral, the system must understand the data series individually and understand how their relationships behave historically. A benchmark constructed without this step risks double-counting redundant information, hard-coding unstable relationships, or mistaking trend correlation for economic structure.

The study has four objectives. First, it audits the currently available U.S. macroeconomic universe and defines a clean monthly analytical subset. Second, it maps contemporaneous co-movement after economically simple transformations. Third, it separates potential information redundancy from merely high Pearson correlation by requiring rank-consistent evidence. Fourth, it tests whether relationships are contemporaneous, temporally shifted, or unstable through time. The output is intended to inform the architecture of the future benchmark object $B(i,t) = \{ \text{center}(i,t), \text{dispersion}(i,t), \text{historical distribution}(i,t) \}$ rather than to estimate that benchmark directly.

$$\text{Benchmark}(i,t) = \{ \text{center}(i,t), \text{dispersion}(i,t), \text{historical distribution}(i,t) \}$$

The empirical scope is the United States. The current repository contains 36 native-frequency FRED series. The relationship notebooks deliberately focus on 27 monthly series; nine weekly or quarterly series are preserved in the resource universe but excluded from the monthly correlation layer to avoid artificial resampling and fabricated observations.

2. Data Architecture, Quality Control, and Variable Universe

The active ingestion design is conservative and offline. The current README specifies that `dataCleansing.py` is the active entry point, while earlier modular ingestion code under `src/chronoz_quant` is retained as legacy work. Raw CSV files are not rewritten or moved. Schema detection, duplicate checks, numeric parsing, missing-value review, date-gap diagnostics, frequency inference, and output generation are performed locally. No interpolation, forward fill, backward fill, normalization, resampling, or economic transformation is performed during cleaning. This boundary is important because it prevents the cleaning stage from silently becoming a modelling stage.

Category	Registered series	Monthly in relationship analysis
Growth / Activity	8	8
Labor	8	7
Inflation	6	6
Credit	5	2
Policy / Liquidity	5	3
External / Structural	4	1
Total	36	27

Table 1. Registered CHronoZ U.S. macroeconomic universe and current monthly analytical subset.

All 36 registered datasets were found and cleaned successfully; no series failed ingestion. Five series generated warnings because one missing observation dated 2025-10-01 was removed from the cleaned training input: Real Retail & Food Services Sales (RRSFS), Unemployment Rate (UNRATE), Labor Force Participation Rate (CIVPART), CPI All Items (CPIAUCSL), and Core CPI (CPILFESL). The raw records remain preserved and no date was filled or shifted. This creates a visible nonconsecutive gap that the notebooks explicitly skip when constructing one-month changes.

The nine excluded series are not rejected economically. Initial Claims, Reserve Balances, and Federal Reserve Total Assets are weekly; the two SLOOS measures, Business Loan Delinquency, Real Exports, Real Imports, and Nonfarm Business Labor Productivity are quarterly. Their exclusion is a frequency-scope decision for this first relationship study, not a statement of irrelevance.

3. Economic Meaning of the Analytical Variables

The analysis uses series identifiers as code keys, but interpretation must remain tied to the real-world quantity represented by each series. The current 27-series monthly subset includes production and consumption, labor utilization, consumer and producer prices, private credit quantities, monetary-policy/liquidity measures, and industrial capacity. The complete 36-series dictionary, including native frequencies and observation histories, is provided in Appendix A.

Code	Economic variable	Real-world representation
INDPRO	Industrial Production	Real output from U.S. manufacturing, mining, and utilities.
IPMAN	Manufacturing Production	Real output specifically from U.S. manufacturing industries.
PCEC96	Real Personal Consumption Expenditure	Inflation-adjusted household spending on goods and services.
PAYEMS	Nonfarm Payroll Employment	Number of payroll jobs at U.S. nonfarm establishments.
UNRATE	Unemployment Rate	Share of the civilian labor force unemployed and actively seeking work.
CES0500000003	Average Hourly Earnings	Average hourly earnings of private nonfarm payroll employees.
CPIAUCSL	CPI All Items	Consumer price level for a broad urban goods-and-services basket.
PCEPI	PCE Price Index	Prices of goods and services in personal consumption expenditures.
BUSLOANS	Commercial & Industrial Loans	Outstanding commercial and industrial loans at U.S. commercial banks.
FEDFUNDS	Federal Funds Rate	Effective overnight interbank reserve-lending rate.
M2SL	M2 Money Supply	Broad money stock including currency, transaction deposits, and liquid savings.
TCU	Total Capacity Utilization	Industrial output as a share of estimated production capacity.

Table 2. Selected variable codes and their economic interpretation. Full universe in Appendix A.

4. Methodology

4.1 Pairwise alignment and transformations

The notebooks join monthly observations by calendar date but do not fill absent observations. Each pair therefore uses its own available history. Raw-level Pearson correlation is retained only as a diagnostic because long-run trends can create spurious association. The primary relationship analyses operate on explicit one-month transformations. Positive level quantities and indexes use percentage change; rates, participation measures, hours, and capacity utilization use first differences. A transformation is computed only when observations are in consecutive calendar months, so a removed monthly observation cannot turn a two-month move into a one-month change.

$$\text{Percentage change: } \Delta\%x(t) = 100 \times [x(t) / x(t-1) - 1]$$

$$\text{First difference: } \Delta x(t) = x(t) - x(t-1)$$

First-difference variables are UNRATE, AWHAEPT, AWHMAN, CIVPART, FEDFUNDS, and TCU. The other 21 monthly variables use one-month percentage changes. Across the monthly universe, 20,728 transformed observations are available. All contemporaneous Pearson and Spearman pair correlations require at least 60 shared observations.

4.2 Correlation and rank robustness

Pearson correlation measures linear co-movement and is sensitive to extreme observations. Spearman correlation measures rank association and is less sensitive to the magnitude of outliers. The analysis uses both because a very large Pearson coefficient accompanied by a near-zero Spearman coefficient is a warning that the apparent relationship may be driven by crisis observations, composition effects, or a small number of extreme monthly moves.

4.3 Redundancy screening

Redundancy analysis is designed to identify pairs that may repeat similar information inside a future aggregate node. A pair becomes a redundancy candidate only when $|\text{Pearson}| \geq 0.80$, $|\text{Spearman}| \geq 0.60$, the two coefficients have matching signs, and the pair has at least 60 observations. This screen does not imply that either variable should be deleted. It identifies where hierarchical aggregation, sub-blocking, or weight control may be required to prevent double-counting.

4.4 Lead-lag scan

For each of 351 monthly pairs, correlation is evaluated over calendar lags from -12 to +12 months, producing 8,775 pair-lag evaluations. The convention is correlation of $X(t)$ with $Y(t+k)$; a positive lag means the first variable precedes the second. Peak-lag selection is descriptive because multiple comparisons, publication timing, revisions, and common shocks can create apparent timing structure. The analysis therefore records zero-lag correlation and the absolute gain of the peak over zero lag rather than treating a nonzero peak as a predictive signal.

$$\rho_{XY}(k) = \text{Corr}[\Delta X(t), \Delta Y(t+k)], \quad k \in \{-12, \dots, +12\}$$

4.5 Rolling relationship stability

The rolling study estimates trailing 60-calendar-month Pearson correlations with a minimum of 48 paired observations and a requirement that the full elapsed window exists. It creates 351 pair paths and 189,286 defined rolling estimates. Because adjacent windows overlap heavily, these estimates are descriptive states of relationship strength rather than independent regime observations.

5. Results and Analysis

5.1 Raw-level versus transformed co-movement

The raw-level matrix is visually dominated by positive relationships among trending level series. This is expected and is precisely why it is treated as diagnostic only. After one-month transformations, the matrix becomes materially sparser and economically more interpretable. All 351 unique monthly pairs have defined Pearson and Spearman coefficients. The transformed overlap ranges from 199 to 1,290 observations; no pair falls below the 60-observation threshold.

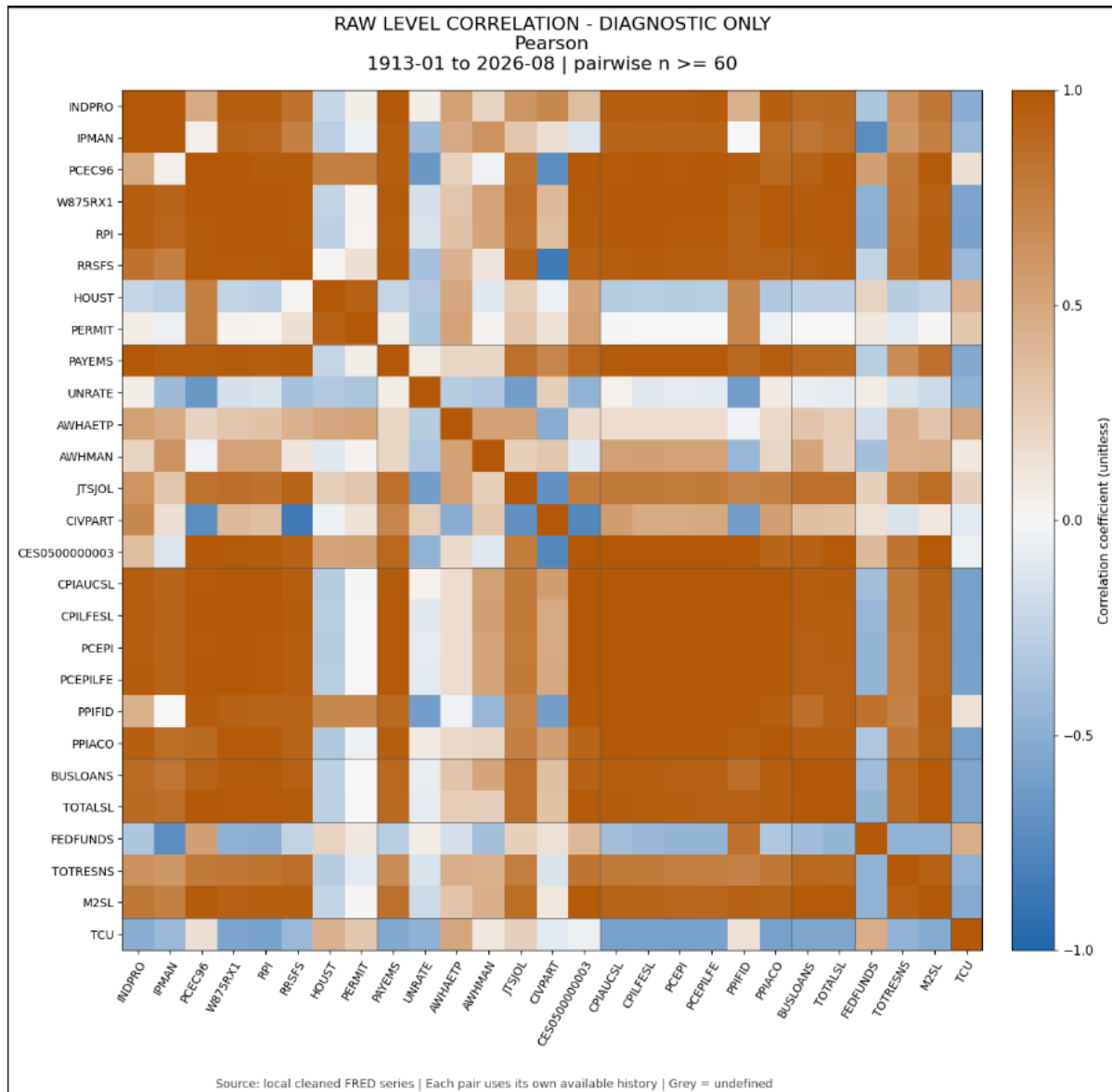


Figure 1. Raw-level Pearson correlation matrix (diagnostic only). Source: `notebook/matrix-corrAnalysis.ipynb`; `outputs/correlation/raw_level_pearson.png`.

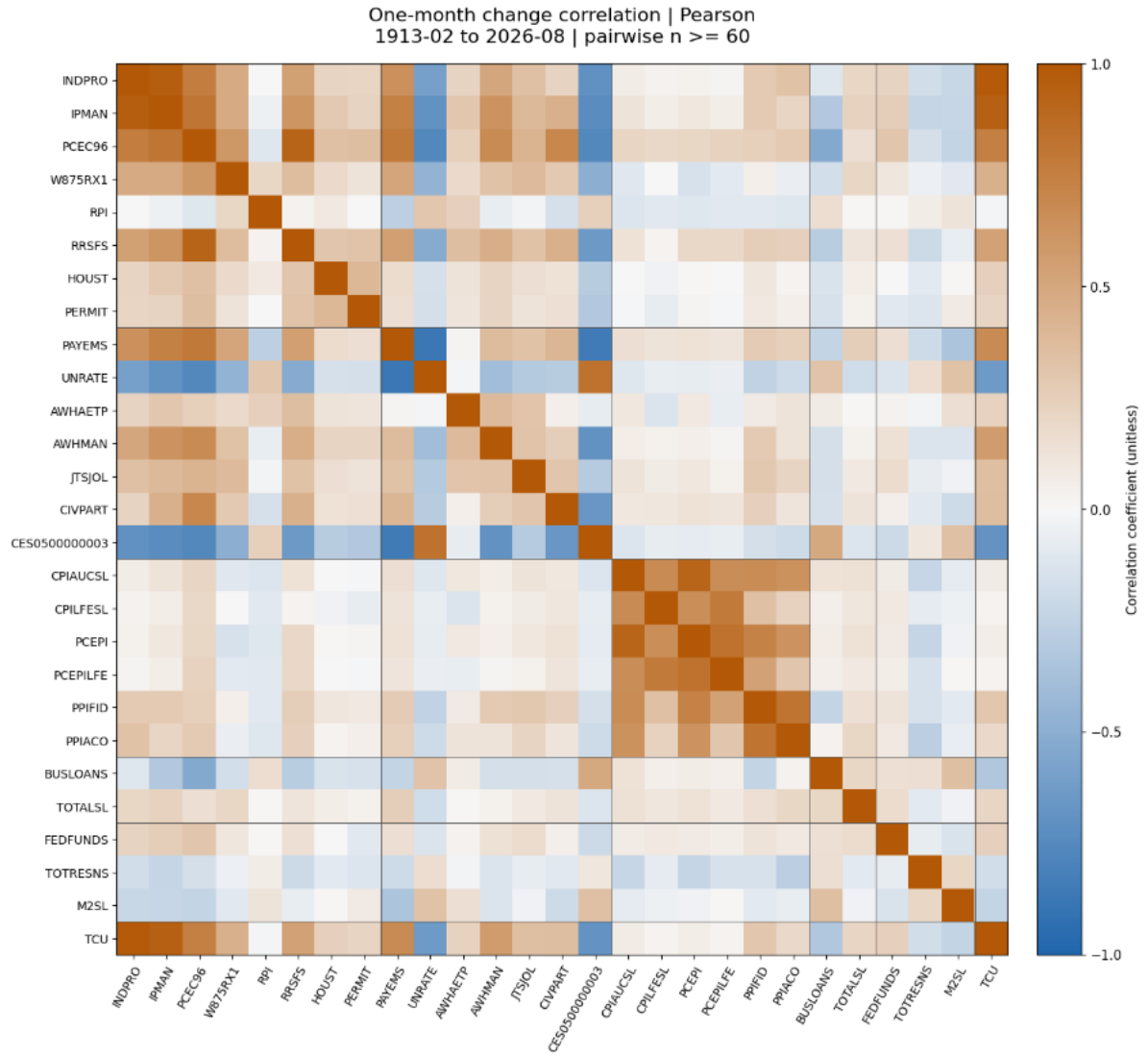


Figure 2. One-month change Pearson correlation matrix. Source: [outputs/correlation/change_pearson.png](#).

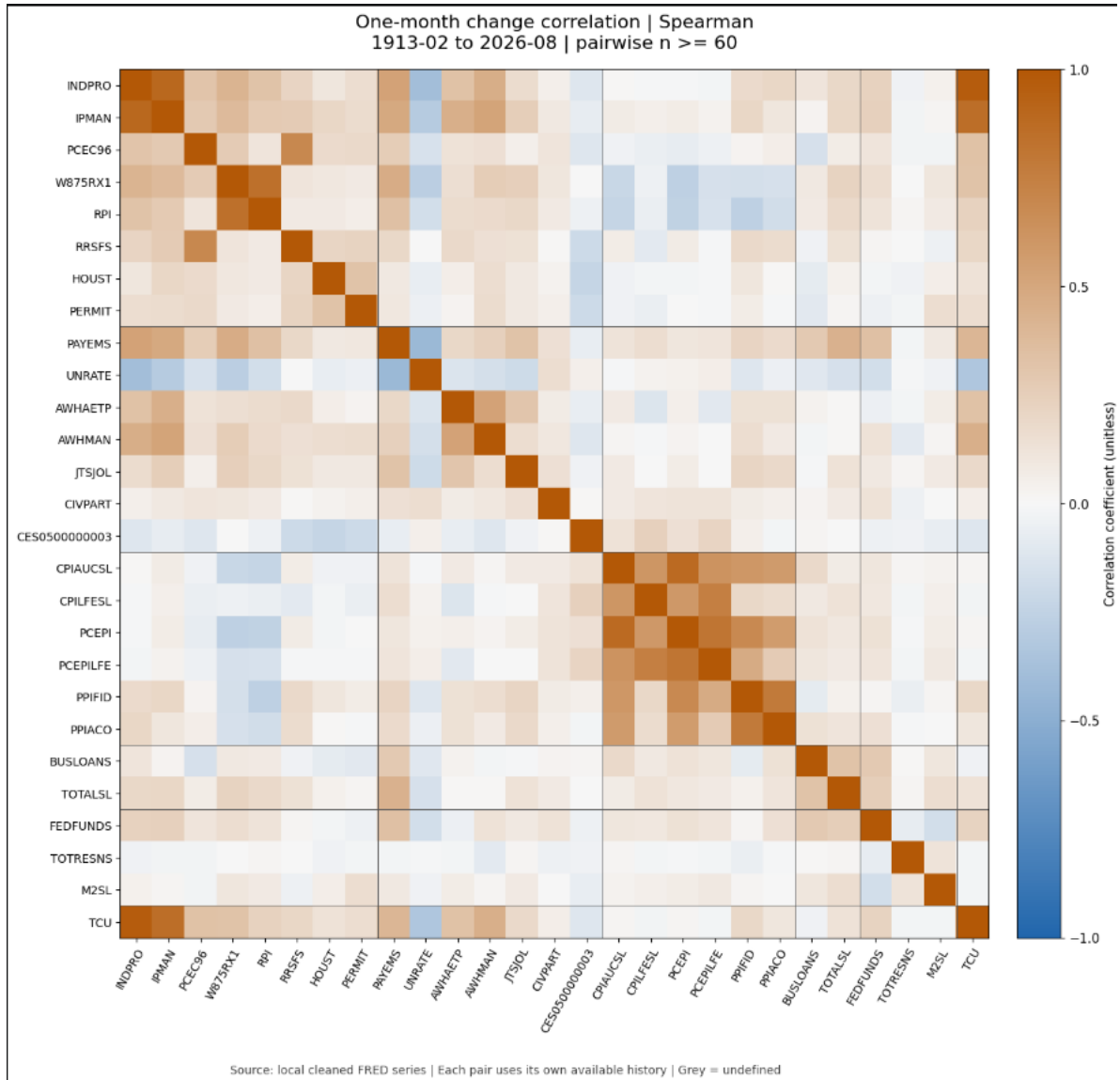


Figure 3. One-month change Spearman correlation matrix. Source: [outputs/correlation/change_spearman.png](#).

Pair	Pearson	Spearman	n	Interpretation
INDPRO - TCU	+0.9837	+0.9625	714	Extremely strong industrial common-state relationship.
INDPRO - IPMAN	+0.9561	+0.8962	654	Strong overlap between total industrial and manufacturing output.
PCEC96 - RRSFS	+0.9213	+0.6880	232	Consumption measures co-move strongly but are not identical.
CPIAUCSL - PCEPI	+0.9113	+0.8826	808	Robust common consumer-inflation factor.
PAYEMS - UNRATE	-0.8786	-0.4281	941	Strong linear opposite movement, but weaker rank stability.
PAYEMS - AHE	-0.8469	-0.0670	245	Pearson is large while rank association is near zero; outlier-sensitive.

Pair	Pearson	Spearman	n	Interpretation
UNRATE - AHE	+0.8290	+0.0535	243	Large Pearson but negligible Spearman; do not treat as stable structure.

Table 3. Selected transformed contemporaneous relationships and robustness interpretation.

The most important robustness lesson is that magnitude alone is insufficient. PAYEMS-Average Hourly Earnings and UNRATE-Average Hourly Earnings have absolute Pearson coefficients above 0.82, yet Spearman coefficients close to zero. These relationships fail the redundancy screen and should be treated as candidates for outlier or crisis-composition diagnostics rather than stable economic links. By contrast, industrial and major inflation pairs retain high rank association and therefore provide stronger evidence of persistent common information.

5.2 Redundancy structure

Seven of the 351 monthly pairs satisfy the joint redundancy screen. They connect 10 series, leaving 17 monthly series unconnected at the selected threshold. Four additional pairs have $|\text{Pearson}| \geq 0.80$ but fail the rank/sign requirement. The resulting network is sparse rather than universal, which is desirable: redundancy is concentrated in identifiable clusters instead of implying that the entire macro universe is one factor.

Series 1	Series 2	Pearson	Spearman	n
INDPRO	TCU	0.9837	0.9625	714
INDPRO	IPMAN	0.9561	0.8962	654
IPMAN	TCU	0.9385	0.8565	654
PCEC96	RRSFS	0.9213	0.6880	232
CPIAUCSL	PCEPI	0.9113	0.8826	808
PCEPI	PCEPILFE	0.8306	0.8116	810
PPIFID	PPIACO	0.8148	0.7793	201

Table 4. Redundancy candidates under the joint Pearson-Spearman screen.

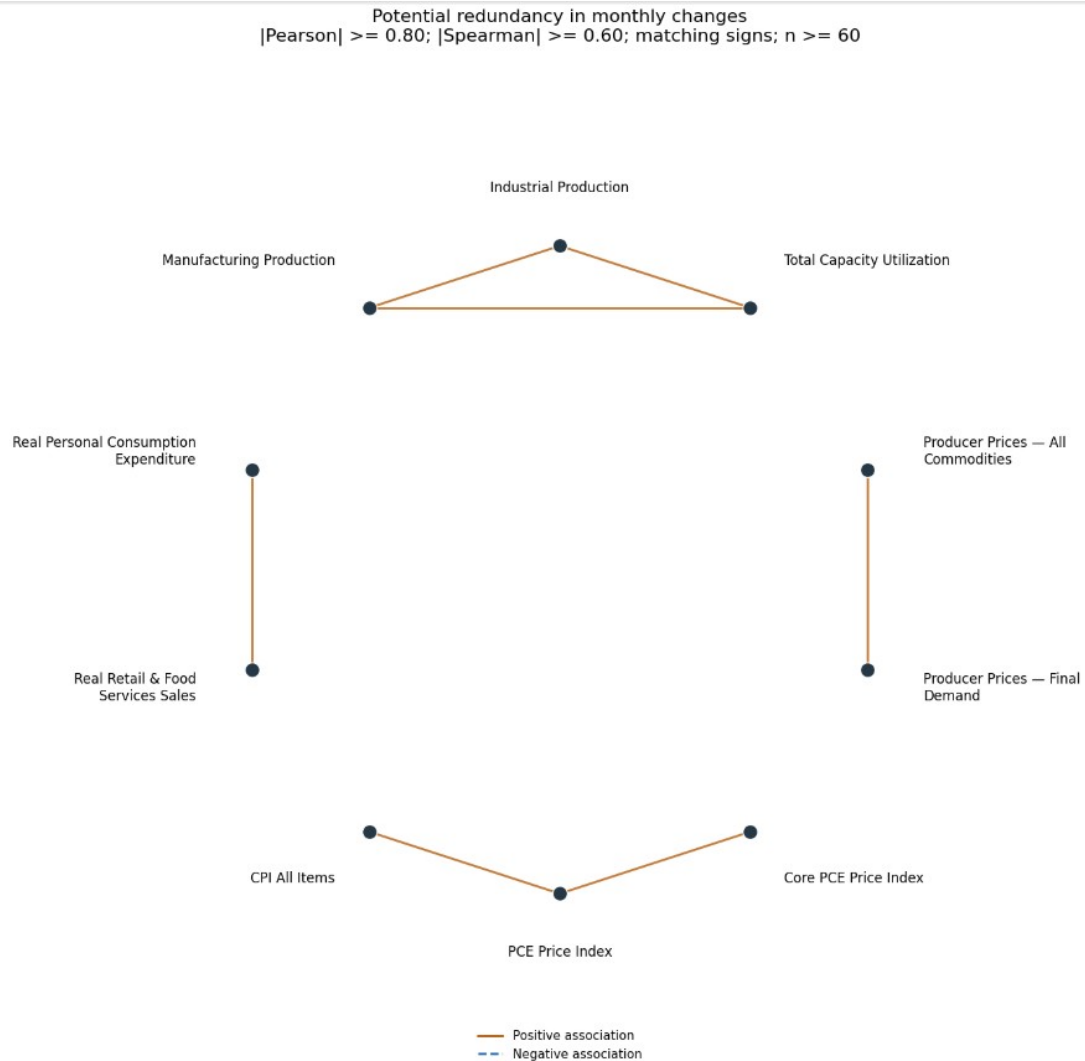


Figure 4. Candidate redundancy network. The industrial triangle is the densest cluster; consumption and inflation pairs form smaller clusters. Source: `notebook/redundancy-map.ipynb`; `outputs/redundancy/redundancy_map.png`.

The industrial cluster (INDPRO, IPMAN, TCU) is the clearest case of repeated information. This does not make the variables useless; it means a future Growth/Activity node should not count three highly overlapping industrial measures as three independent votes. PCEC96 and RRSFS form a consumption cluster. Inflation separates into a consumer-price cluster and a producer-price cluster, which is economically useful because producer-price pressure is not interchangeable with consumer-price inflation.

The redundancy result also provides an ontology correction signal. TCU is currently registered under External/Structural, but both its economic meaning and its empirical clustering place it much closer to industrial activity. Reclassification should be considered at the modelling stage while retaining the raw series identifier and provenance.

5.3 Lead-lag structure

The lag scan evaluates 8,775 pair-lag combinations with no evaluation falling below the minimum-overlap rule. Of 351 pairs, 194 (55.3%) achieve their maximum absolute Pearson correlation at zero lag and 157 (44.7%) peak away from zero. This does not mean that 44.7% of relationships are predictive. For many pairs, the gain over zero lag is small, and peak selection itself searches across 25 alternatives. The strongest pairs in the system remain overwhelmingly zero-lag relationships.

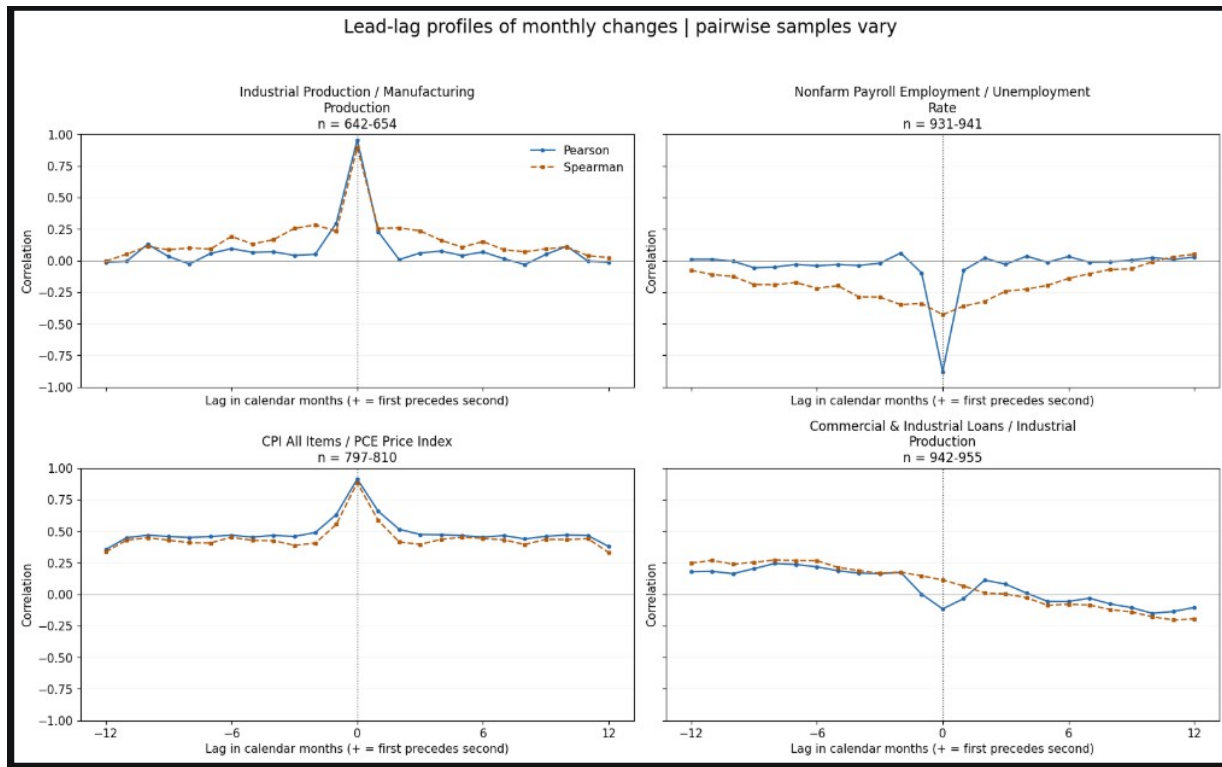


Figure 5. Lead-lag profiles for selected monthly-change pairs. Strong industrial, inflation, and payroll-unemployment relationships peak at lag 0, while C&I Loans-Industrial Production shows weaker state-dependent timing structure.
Source: `notebook/lead-lag-correlation.ipynb`; `outputs/lead_lag/lag_profiles.png`.

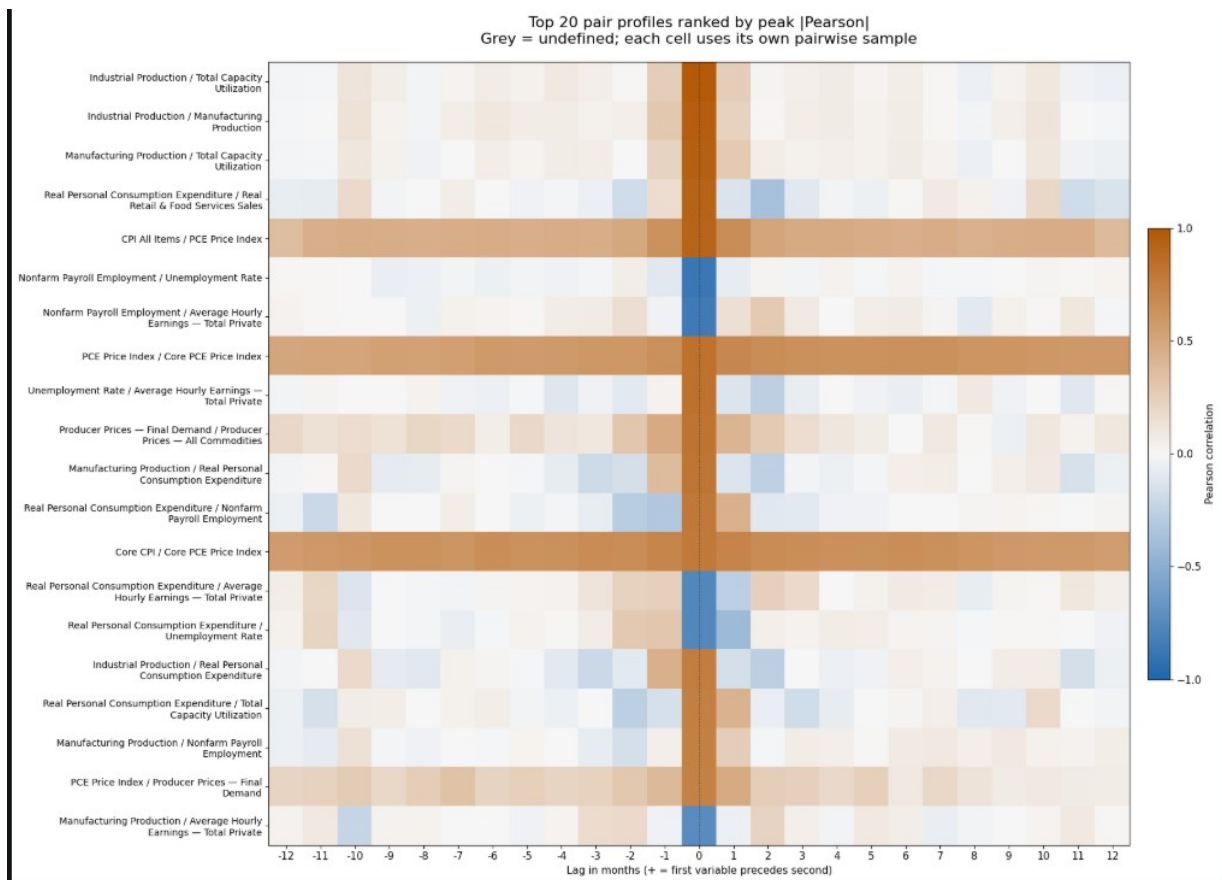


Figure 6. Top pair profiles across -12 to +12 calendar months. The zero-lag column dominates many of the highest- $|\text{Pearson}|$ pairs. Source: `outputs/lead_lag/top_lag_profiles.png`.

The lead-lag evidence supports a distinction between common-state relationships and timing candidates. INDPRO-IPMAN, INDPRO-TCU, CPI-PCEPI, PCEPI-Core PCE, and PAYEMS-UNRATE are primarily same-month relationships. They are useful for state confirmation or

redundancy control, not for a directional transmission claim. A relationship should become a timing candidate only when a nonzero peak has a material gain over zero lag and remains stable under rank, subperiod, and release-timing diagnostics.

5.4 Rolling relationship stability

Rolling analysis is the strongest evidence that a single full-history correlation is not sufficient for CHronoZ. The 60-month paths reveal three qualitatively different relationship classes: highly stable structural overlap, persistent but time-varying relationships, and regime-dependent relationships that can change sign.

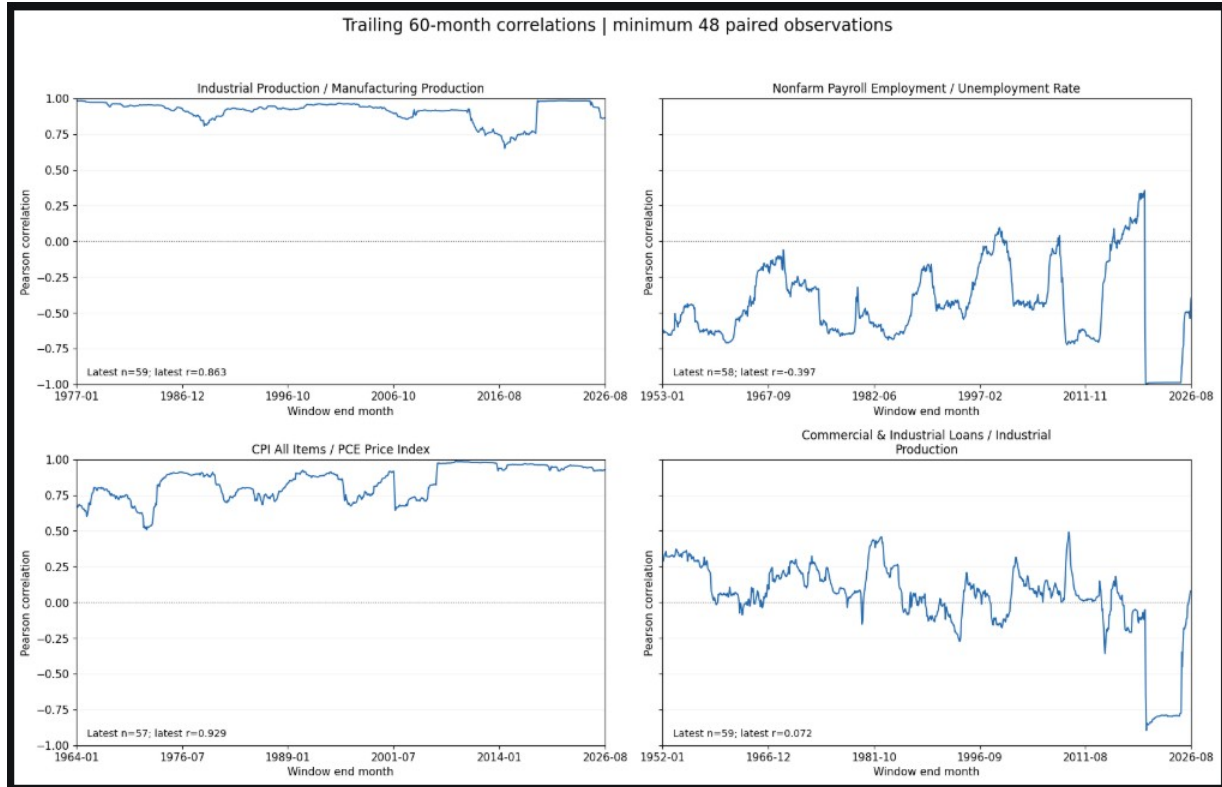


Figure 7. Trailing 60-month Pearson paths for selected pairs. Industrial Production-Manufacturing Production and CPI-PCE remain high, while Payroll-Unemployment and C&I Loans-Industrial Production vary materially through time.

Source: notebook/rolling-correlation.ipynb; outputs/rolling_correlation/rolling_profiles.png.

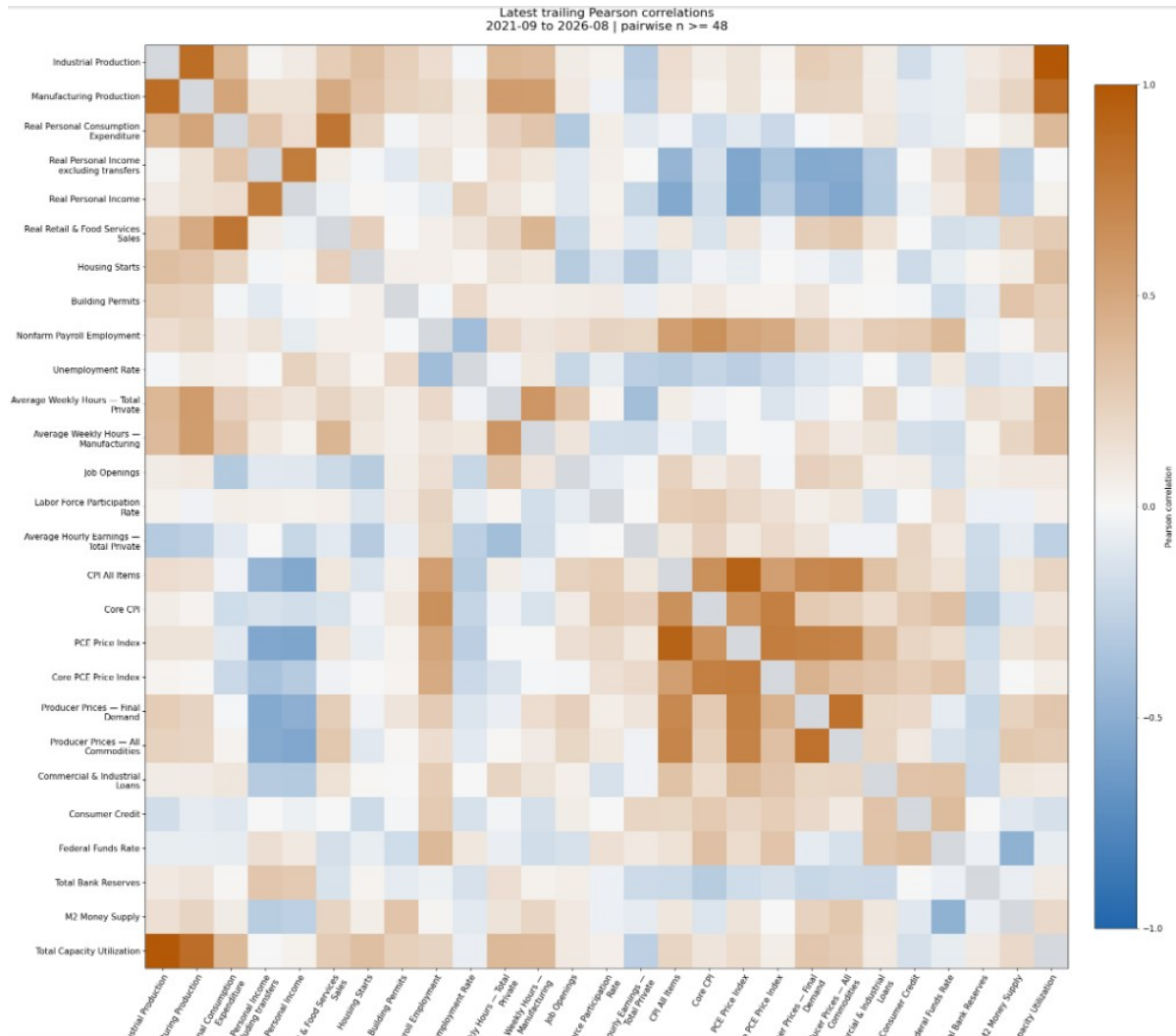


Figure 8. Latest trailing 60-month correlation matrix for 2021-09 to 2026-08. Source: [outputs/rolling_correlation/latest_rolling_matrix.png](#).

Pair	Latest 60m	Min	Max	Range	Assessment
INDPRO - TCU	0.9954	0.9357	0.9992	0.0636	Structural / highly stable
INDPRO - IPMAN	0.8628	0.6499	0.9855	0.3356	Stable structural overlap
PCEC96 - RRSFS	0.8099	0.5081	0.9825	0.4744	Stable but varying consumption link
PAYEMS - UNRATE	-0.397	≈ -1.00	positive episodes	large	Time-varying labor relationship
INDPRO - BUSLOANS	0.0719	-0.8969	0.4920	1.3888	Highly regime-dependent
W875RX1 - RPI	—	-0.8095	0.9982	1.8077	Largest observed range; definition/composition sensitive

Table 5. Selected rolling-stability findings. Approximate PAYEMS-UNRATE extrema are read from the plotted path; other values are from exported stability tables.

Industrial Production-Capacity Utilization is the clearest structural relationship: its rolling correlation remains above approximately 0.94 and is 0.995 in the latest window. Industrial Production-Manufacturing Production is also persistently high. In contrast, Commercial & Industrial Loans-Industrial Production has ranged from strongly negative to moderately positive and is near zero in the latest window. Such a pair should not be hard-coded as a stable transmission relationship. Payroll-Unemployment remains economically related but its rolling strength changes substantially, demonstrating that even intuitive macro relationships can be state-dependent in observed monthly changes.

6. Integrated Findings for CHronoZ Benchmark Architecture

Taken together, the four analyses imply that CHronoZ should not begin from a single aggregated economic score or a fixed causal graph. The empirical evidence supports a layered representation in which each variable owns its benchmark and current state, relationship diagnostics are maintained separately, and aggregation occurs only after redundancy and stability are understood.

Layer	Canonical object	Role in future system
1. Variable	VariableState(i,t)	Transformed value, benchmark distance, percentile, momentum, freshness, provenance.
2. Relationship	RelationshipState(i,j,t)	Current correlation, historical stability, lead-lag diagnostics, redundancy flag, anomaly status.
3. Sub-block	Empirical cluster / economic block	Compress overlapping information without deleting underlying evidence.
4. Macro node	Growth, Inflation, Labor, Credit, etc.	Human-interpretable summary built from controlled sub-blocks.
5. Regime / analogue	State configuration through time	Retrieve historical states and test conditional patterns.
6. Pricing / decision	Separate market-pricing object	Compare economic evidence with expectations and asset prices downstream.

Table 6. Proposed evidence hierarchy implied by the empirical relationship study.

6.1 Implications for benchmark weighting

Redundancy should affect aggregation, not data retention. If INDPRO, IPMAN, and TCU all improve together, CHronoZ should recognize broad industrial confirmation without awarding the industrial theme three times the weight of consumption or housing. A transparent initial architecture is therefore Variable -> Redundancy Cluster -> Sub-block -> Macro Node. Statistical clusters should be compared with economic meaning before they influence weights.

6.2 Relationship anomaly as research evidence

A major opportunity is to benchmark relationships themselves. If two variables have a historically stable relationship and the current rolling relationship deviates sharply, the break can be treated as an anomaly requiring explanation. Conversely, a historically unstable pair should not produce a high-confidence anomaly alert merely because its current correlation changed. A future RelationshipState may therefore include current rolling correlation, historical center and dispersion, sign stability, peak lag, lead-lag gain, sample size, and a structural-versus-regime-dependent classification.

Conceptual relationship anomaly: $A(i,j,t) = \text{current relationship}(i,j,t) - \text{expected historical relationship}(i,j)$

6.3 Implications for machine learning

For machine learning, the current work argues against feeding a large set of raw, highly overlapping macro series into an unconstrained model and asking it to discover everything. Useful engineered features include variable benchmark states, redundancy-cluster summaries, rolling relationship states, relationship anomalies, and carefully validated lead-lag candidates. Walk-forward evaluation must preserve release timing and data vintages. Feature selection should be judged out of sample and should not retroactively define the economic benchmark solely by asset-return Sharpe.

6.4 Implications for an LLM macro agent

The LLM should receive numeric facts rather than manufacture them. A deployable agent contract should expose each variable by human-readable economic name as well as code, its benchmark metadata, current state, transformation, freshness, and provenance. Relationship objects should state whether a link is contemporaneous, redundant, a timing candidate, or regime-dependent. This prevents the agent from converting correlation into causality through fluent narrative. When a normally coherent cluster fragments, the agent can use the anomaly as a research trigger: identify what else was happening in comparable historical episodes, inspect policy and credit conditions, retrieve external evidence, and then compare the resulting economic interpretation with market pricing.

6.5 Implications for trading and decision making

The relationship layer is potentially valuable to a discretionary or systematic trader because the information content may lie in disagreement rather than direction alone. For example, if production and capacity utilization normally co-move but one improves while the other stalls, the divergence may change how broad or narrow an activity improvement should be interpreted. If growth weakness occurs while credit stress remains contained, the asset-premium implications differ from a broad deterioration in which production, labor, and credit weaken together. CHronoZ should therefore use relationship anomalies to change research priority and confidence, not to emit an automatic trade. Economic reality, consensus expectations, and market pricing remain separate downstream objects.

7. Limitations

- The relationship notebooks currently analyze only the 27 monthly series. Weekly and quarterly information remains outside the correlation system rather than being handled with a mixed-frequency method.
- The analysis uses observation dates, not publication timestamps. A statistical lead can disappear or reverse when release timing is modelled correctly.
- The data are current cleaned series, not historical real-time vintages. Revised-data leakage has not yet been addressed for backtesting.
- The one-month transformation registry is economically reasonable for exploration but has not yet been optimized or validated variable by variable against distributional and stationarity diagnostics.
- The lead-lag scan performs many comparisons. Peak correlations require multiple-testing control, subperiod stability, and out-of-sample evaluation before predictive use.
- Rolling windows overlap heavily, so 189,286 defined rolling estimates are not independent observations.
- Correlation, rank association, timing, and redundancy do not establish causal mechanisms. Causal interpretation requires economic mechanisms, temporal fit, alternative explanations, and falsification tests.
- The current category ontology is provisional. TCU is one clear candidate for reclassification from External/Structural to industrial Growth/Activity based on both semantics and empirical clustering.

8. Conclusion

This research establishes a reproducible empirical map of the current CHronoZ U.S. macroeconomic dataset before benchmark calibration or machine learning. The data pipeline successfully preserves and cleans 36 FRED series, while the first relationship layer uses 27 native monthly series without artificial frequency conversion. Transformed contemporaneous correlations show clear common factors in industrial activity, consumption, and inflation, while rank diagnostics expose several apparently strong Pearson relationships as outlier-sensitive. Seven redundancy candidates survive a joint Pearson-Spearman screen, demonstrating that information overlap is concentrated rather than universal.

Lead-lag analysis shows that timing structure exists but that the strongest relationships are mostly contemporaneous: 194 of 351 pairs peak at zero lag, and 157 peak away from zero. Nonzero peaks remain descriptive because release timing and multiple comparisons are not yet controlled. Rolling analysis provides the most important structural result: relationship stability is heterogeneous. Industrial output and capacity utilization behave like a persistent common factor, whereas credit-output and several labor relationships vary materially across time. A single full-history coefficient therefore cannot be treated as permanent economic law.

The evidence supports a benchmark architecture in which variable states remain canonical, redundancy is controlled through hierarchical aggregation, and relationship state is maintained as an independent evidence layer. This structure is compatible with later historical-analogue retrieval, regime modelling, machine learning, LLM-based causal research, and asset-pricing interpretation while preserving the separation between measurement, explanation, and decision.

9. Recommended Next Research Stage

The following work is recommended as the next stage of the CHronoZ benchmark programme. These items are forward research recommendations, not conclusions from the current empirical results.

- Build a per-variable distribution notebook before aggregation: histogram/empirical CDF, quantiles, skewness, kurtosis, mean/median, standard deviation, MAD, seasonality diagnostics, structural-break checks, and stationarity diagnostics.
- Calibrate the benchmark object separately for each variable: center, dispersion, and historical distribution. Compare classical z-score, robust z-score, and empirical percentile instead of imposing one normalization method globally.
- Add real-time vintage and release-calendar infrastructure (for example ALFRED-style vintages) before any walk-forward macro or asset-return backtest.
- Extend relationship diagnostics with subperiod and rolling lead-lag stability. A timing candidate should require meaningful gain over zero lag, sufficient sample size, rank support, and stability across historical subperiods.
- Apply multiple-testing controls and dependence-aware uncertainty methods to lead-lag discovery, including false-discovery-rate procedures and block-bootstrap confidence intervals where practical.
- Create a RelationshipState schema containing pair identifiers, economic names, current/long-run/rolling association, sign stability, peak lag, lead-lag gain, redundancy class, sample size, and provenance.
- Revisit the economic ontology after empirical review. In particular, evaluate moving TCU into an Industrial Activity sub-block and use redundancy clusters to design sub-block weights.
- Introduce weekly and quarterly data through explicit mixed-frequency logic rather than naive forward filling. Preserve native frequency as metadata and make cross-frequency alignment an explicit modelling decision.
- Use PCA/factor models only as comparative compression tools after the transparent benchmark is working. Compare factor weights against hierarchical equal-weight structures out of sample.

- Build historical-analogue retrieval on stored VariableState and RelationshipState snapshots so the agent can ask: when did the current configuration or anomaly occur before, what else was happening, and what followed?
- Keep the future PricingState separate from EconomicState. Only after economic measurement is stable should CHronoZ compare reality, consensus, market pricing, and subsequent asset returns.
- Define a strict machine-readable JSON/API contract so the LLM can consume reproducible quantitative outputs without recreating benchmark mathematics in prose.

References and Reproducibility Sources

- [1] Rizuqura. CHronoZ-economicBenchmarkModelling. GitHub repository, commit e2153e19ea20e9a3675f0dc27d6f468458731acc, accessed 18 September 2026.
<https://github.com/Rizuqura/CHronoZ-economicBenchmarkModelling>
- [2] Federal Reserve Bank of St. Louis. Federal Reserve Economic Data (FRED). Source used by the repository data registry and cleaned-series catalog.
- [3] QuantFun x Wonnyy. CHronoZ - Macro Intelligence Agent: Empirical Economic Benchmark, State, Pricing & Decision Architecture. Whitepaper v0.2, 16 September 2026.
- [4] Pearson, K. (1895). Notes on regression and inheritance in the case of two parents. Proceedings of the Royal Society of London, 58, 240-242.
- [5] Spearman, C. (1904). The proof and measurement of association between two things. American Journal of Psychology, 15(1), 72-101.

Appendix A. CHronoZ U.S. Macroeconomic Variable Dictionary

This appendix preserves the current registered 36-series universe. The monthly relationship notebooks use only the monthly subset; weekly and quarterly series remain available for later mixed-frequency analysis. Observation histories are taken from the repository resource catalog generated on 17 September 2026.

Category	Code	Variable	Real-world representation	Freq.	Start	End	Obs.
Growth / Activity	INDPRO	Industrial Production Index	Real output produced by U.S. manufacturing, mining, and utilities.	M	1919-01	2026-07	1291
Growth / Activity	IPMAN	Industrial Production: Manufacturing	Real output produced by U.S. manufacturing industries.	M	1972-01	2026-07	655
Growth / Activity	PCEC96	Real Personal Consumption Expenditures	Inflation-adjusted spending by U.S. consumers on goods and services.	M	2007-01	2026-07	235
Growth / Activity	W875RX1	Real Personal Income ex Transfers	Inflation-adjusted personal income excluding current transfer receipts.	M	1959-01	2026-07	811
Growth / Activity	RPI	Real Personal Income	Inflation-adjusted personal income including earnings, property income, and transfers.	M	1959-01	2026-07	811
Growth / Activity	RRSFS	Real Retail & Food Services Sales	Inflation-adjusted retail and food-service sales.	M	1992-01	2026-08	415
Growth / Activity	HOUST	Housing Starts	New privately owned housing units on which construction has started.	M	1959-01	2026-08	812
Growth / Activity	PERMIT	Building Permits	New privately owned housing units authorized by permits.	M	1960-01	2026-08	800
Labor	PAYEMS	Nonfarm Payroll Employment	Payroll jobs at U.S. nonfarm establishments; jobs rather than unique people.	M	1939-01	2026-08	1052
Labor	UNRATE	Unemployment Rate	Percentage of the civilian labor force unemployed and actively seeking work.	M	1948-01	2026-08	943
Labor	ICSA	Initial Claims	Initial unemployment-	W	1967-01	2026-09	3115

Category	Code	Variable	Real-world representation	Freq.	Start	End	Obs.
			insurance applications; a weekly measure of new claims.				
Labor	AWHAETP	Average Weekly Hours: Total Private	Average weekly hours worked by employees on private nonfarm payrolls.	M	2006-03	2026-08	246
Labor	AWHMAN	Average Weekly Hours: Manufacturing	Average weekly hours worked by production/nonsupervisory manufacturing employees.	M	1939-01	2026-08	1052
Labor	JTSJOL	Job Openings: Total Nonfarm	Number of unfilled job openings at nonfarm establishments.	M	2000-12	2026-07	308
Labor	CIVPART	Labor Force Participation Rate	Share of civilian population age 16+ working or actively seeking work.	M	1948-01	2026-08	943
Labor	CES0500000003	Average Hourly Earnings: Total Private	Average hourly earnings of employees on private nonfarm payrolls.	M	2006-03	2026-08	246
Inflation	CPIAUCSL	CPI All Items	Prices paid by urban consumers for a broad goods-and-services basket.	M	1947-01	2026-08	955
Inflation	CPILFESL	Core CPI	Urban consumer prices excluding food and energy.	M	1957-01	2026-08	835
Inflation	PCEPI	PCE Price Index	Prices of goods and services included in personal consumption expenditures.	M	1959-01	2026-07	811
Inflation	PCEPILFE	Core PCE Price Index	PCE prices excluding food and energy.	M	1959-01	2026-07	811
Inflation	PPIFID	Producer Prices - Final Demand	Selling prices received by domestic producers for final-demand output.	M	2009-11	2026-08	202
Inflation	PPIACO	Producer Prices - All Commodities	Selling prices received by domestic producers for a broad commodity set.	M	1913-01	2026-08	1364
Credit	BUSLOANS	Commercial & Industrial	Outstanding commercial	M	1947-01	2026-08	956

Category	Code	Variable	Real-world representation	Freq.	Start	End	Obs.
		Loans	and industrial loans held by U.S. commercial banks.				
Credit	TOTALSL	Consumer Credit	Outstanding consumer credit, revolving and nonrevolving, excluding home mortgages.	M	1943-01	2026-07	1003
Credit	DRTSCILM	SLOOS: Tightening C&I Standards	Net share of surveyed banks reporting tighter C&I lending standards.	Q	1990-04	2026-07	146
Credit	DRSDCILM	SLOOS: Stronger C&I Loan Demand	Net share of surveyed banks reporting stronger C&I loan demand.	Q	1991-10	2026-07	140
Credit	DRBLACBS	Business Loan Delinquency Rate	Percentage of business loans at commercial banks that are delinquent.	Q	1987-01	2026-04	158
Policy / Liquidity	FEDFUNDS	Federal Funds Effective Rate	Effective overnight rate for reserve-balance lending between depository institutions.	M	1954-07	2026-08	866
Policy / Liquidity	TOTRESNS	Total Bank Reserves	Total reserves of depository institutions, including reserve balances and qualifying vault cash.	M	1959-01	2026-07	811
Policy / Liquidity	WRESBAL	Reserve Balances with Federal Reserve Banks	Reserve balances held by depository institutions at Federal Reserve Banks.	W	2002-12	2026-09	1239
Policy / Liquidity	WALCL	Federal Reserve Total Assets	Total assets on the Federal Reserve balance sheet.	W	2002-12	2026-09	1239
Policy / Liquidity	M2SL	M2 Money Stock	Broad money stock: currency, transaction deposits, and selected liquid savings instruments.	M	1959-01	2026-07	811
External / Structural	EXPGSC1	Real Exports of Goods and Services	Inflation-adjusted value of U.S. exports of goods and services.	Q	1947-01	2026-04	318
External / Structural	IMPGSC1	Real Imports of Goods and Services	Inflation-adjusted value of U.S. imports of goods and	Q	1947-01	2026-04	318

Category	Code	Variable	Real-world representation	Freq.	Start	End	Obs.
			services.				
External / Structural	OPHNFB	Nonfarm Business Labor Productivity	Real output per hour worked in the nonfarm business sector.	Q	1947-01	2026-04	318
External / Structural	TCU	Capacity Utilization: Total Industry	Industrial output as a percentage of estimated production capacity.	M	1967-01	2026-07	715

Appendix Table A1. Current registered variable universe. M = monthly, W = weekly, Q = quarterly. Source: Resources.md and data/cleaned/series_metadata.csv in repository snapshot.